

Evolutionary dynamics-aware *de novo* design of a thermostable NS1-derived diagnostic protein for Yellow Fever virus

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Supporting information - Graphics

Figure S1

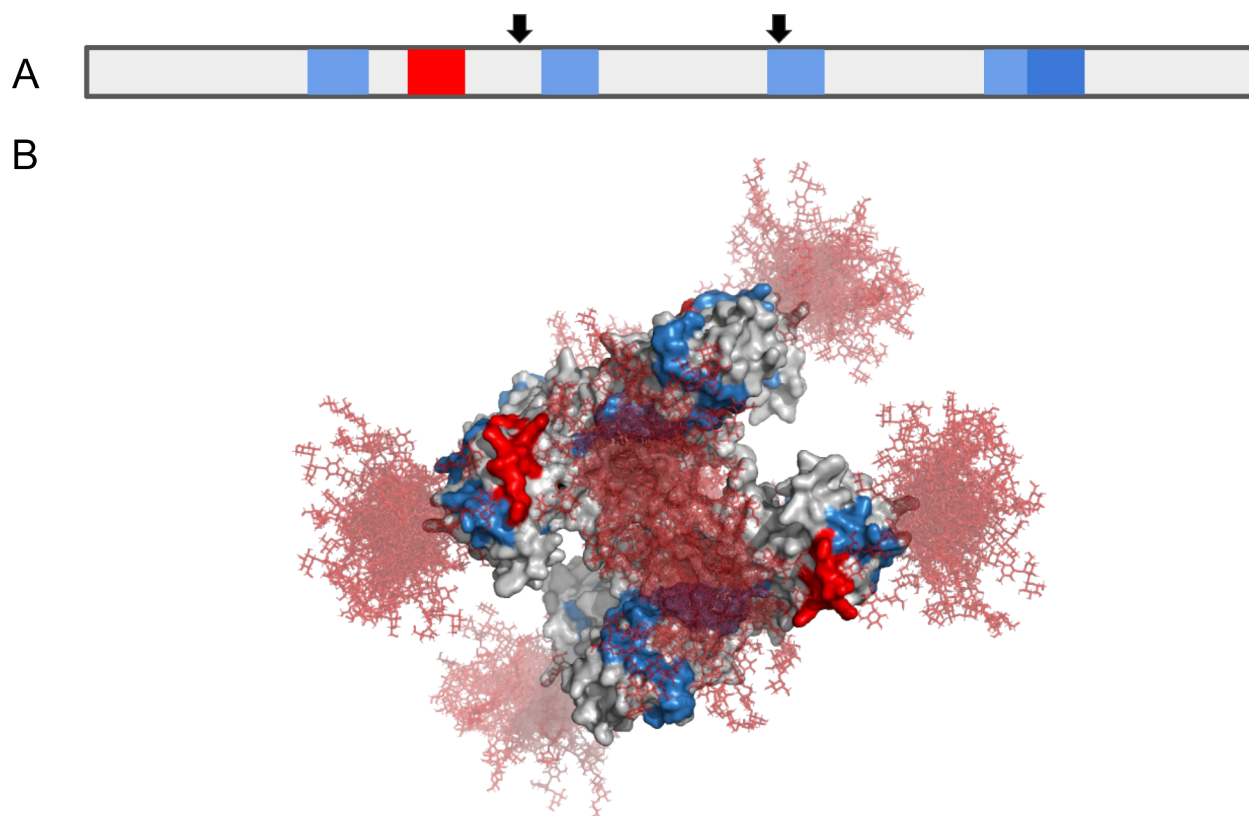


Figure S1. Unique regions mapped onto the YFV NS1 tetramer (PDB: 8ZBA). (a) Schematic representation of the NS1 protein sequence. The region selected for downstream analyses is highlighted in red, whereas the remaining mapped regions are shown in blue. Predicted glycosylation sites are indicated by arrows. (b) Glycan conformers generated using GlycoSHIELD are shown in light red.

Figure S2

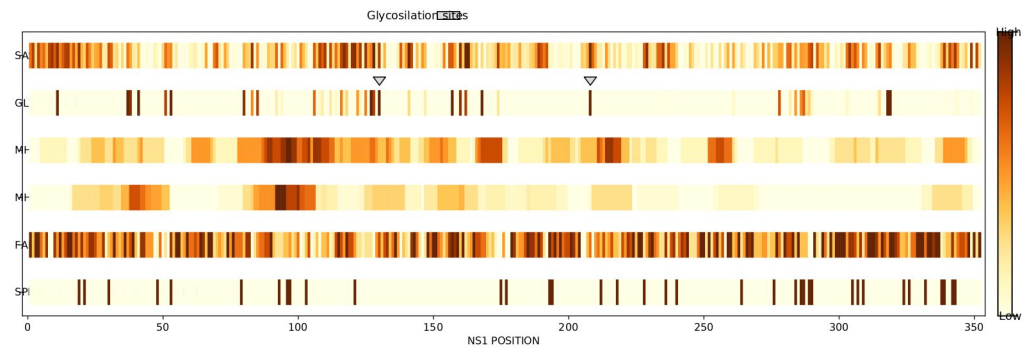


Figure S2. Evolutionary and structural properties of the native NS1 protein. Each row represents a different feature, including relative solvent-accessible surface area (SASA RELATIVE), steric occlusion by glycan conformers (GLYCAN SHIELDING), number of predicted major histocompatibility complex (MHC) class I binding epitopes (MHC-I BINDERS), number of predicted MHC class II binding epitopes (MHC-II BINDERS), evolutionary conservation inferred using ConSurf (FAMILY-LEVEL CONSERVATION), and amino acid substitutions inferred through ancestral character reconstruction across the YFV phylogeny (SPECIES-LEVEL CHANGES OVER TIME). Grey arrows indicate known glycosylation sites.

Figure S3

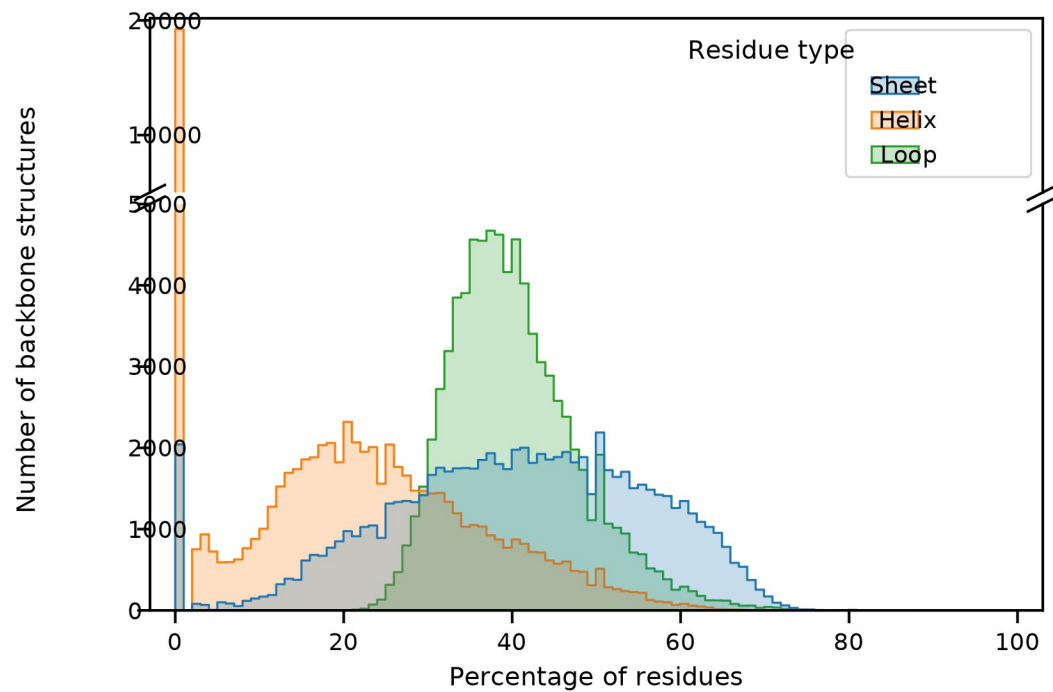


Figure S3. Secondary structure composition of the generated backbones scaffolds. Histogram showing the proportion of residues assigned as loop (green), α -helix (orange) or β -sheet (blue) secondary structure elements across all backbones containing the selected YFV NS1 peptide.

Figure S4

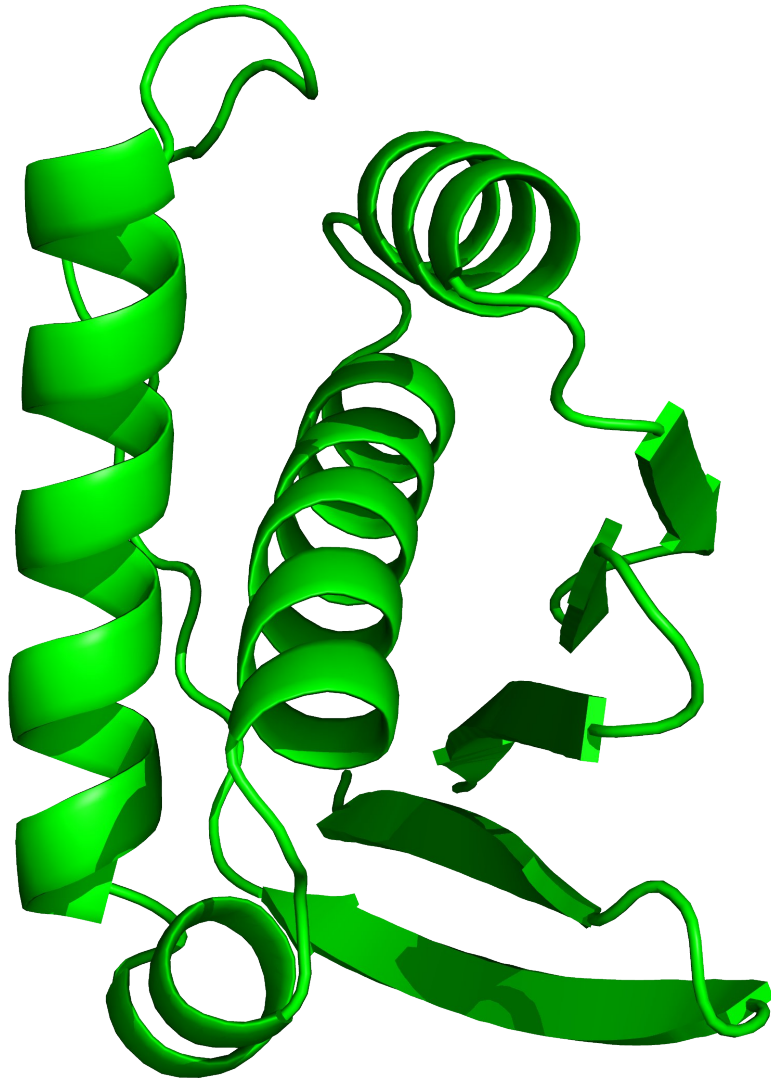


Figure S4. YFV-Scaffold structure.

Figure S5

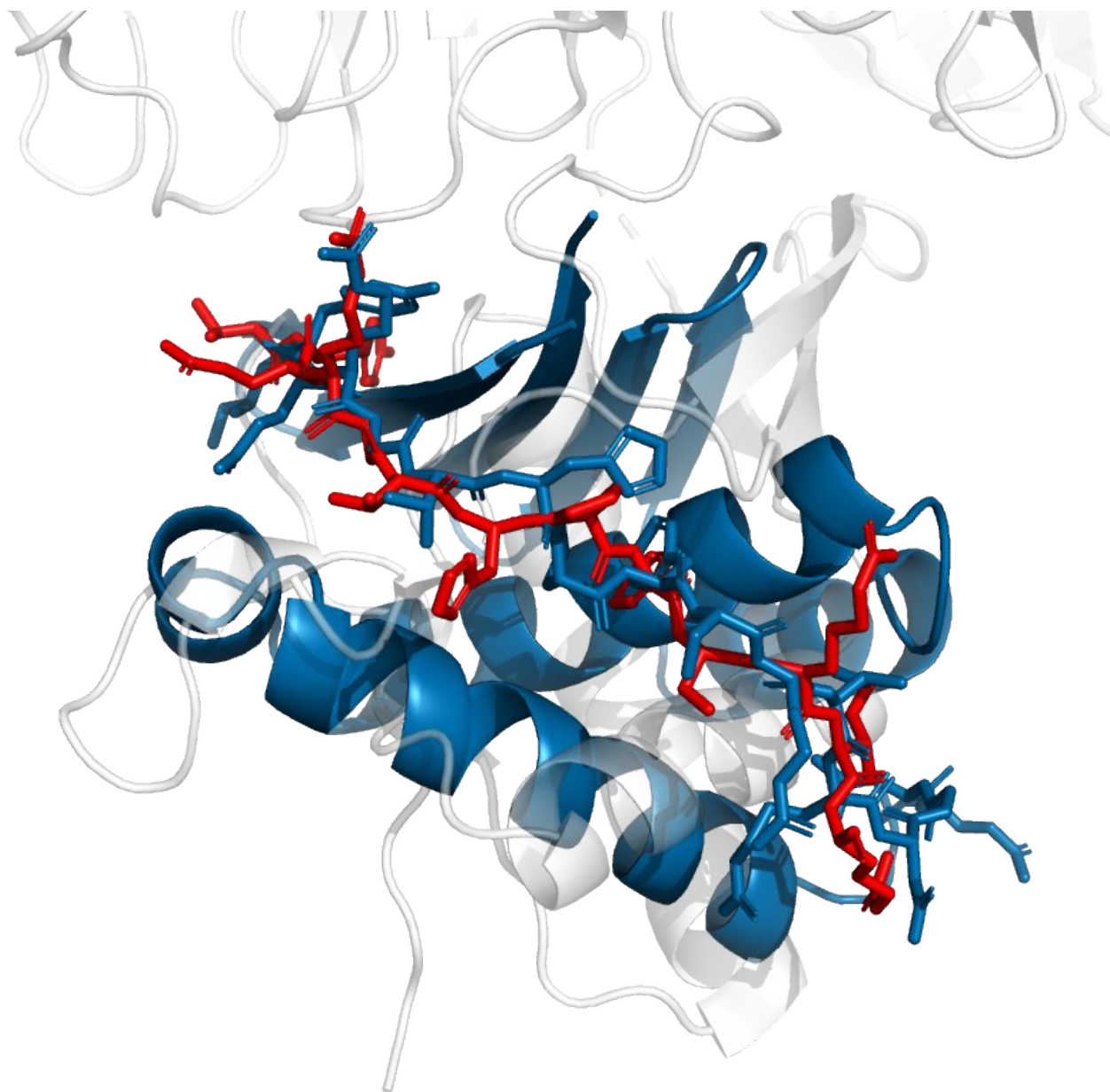


Figure S5. Structural superposition of YFV-Scaffold and the native YFV NS1 protein. The designed protein (YFV-Scaffold, blue) is superimposed onto the crystal structure of YFV NS1 (PDB ID: 8ZBA, white). Residues corresponding to the native epitope incorporated into the YFV-Scaffold design are highlighted in red.